Exploratory Data Analysis Report: Titanic Dataset Insights & Pipeline Integration

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Course: ITAI 1371 – Principles of Machine Learning

Dataset: Titanic Passenger Survival Records (891 rows)

Arinze Okafor

1. Overview & Objectives

Exploratory Data Analysis (EDA) audits raw data to reveal distributions, spot missing values, and check feature correlations before model training.

2. Statistical Summaries vs. Visual Findings

- **Overall Baseline:** 38.4% survival rate (class imbalance). Ages span 0.42 to 80 years (avg 29.7). Fares range from $0 to $512.33 (median $14.45), showing heavy right-skewness.
- **Key Visual Trends:**
    - **Class:** 1st Class had much higher survival than 3rd Class.
    - **Gender:** Over 70% of females survived vs. under 20% of males.
    - **Age:** High mortality for ages 20–40, but a clear survival spike for children under 5.
    - **Port:** Cherbourg boarders had higher survival, tied to more 1st Class passengers.
    - **Fare:** Log-scaled boxplots confirm survivors paid higher median fares.

3. Data Preprocessing Challenges

- **Missing Values:** Age (19.8% missing) needs median group imputation. Cabin (77% missing) should be dropped or turned into a binary flag (Has_Cabin). Embarked (2 missing) fills via mode.

- **Feature Skewness:** Fare outliers require log transformation ($\log(\text{Fare} + 1)$) or scaling to protect linear models.
- **Encoding:** Sex and Embarked require One-Hot or binary encoding.

4. Pipeline Integration

- **Engineering:** Create composite features like Is_Child or Family_Size.
- **Algorithms:** Tree models naturally capture non-linear trends (like child survival spikes); linear models require scaled inputs.
- **Metrics:** Imbalanced classes require evaluating Precision, Recall, and F1-Score over raw Accuracy.

Hamza Ajmal

Reflective Journal: Module 04 Lab, Exploratory Data Analysis

ITAI 1371 - Introduction to Machine Learning

How my view of EDA changed

Going in, I thought EDA meant making a handful of charts to decorate a notebook. By the end I saw it as the step where you form questions about the data before any model exists. Each plot in the lab acted like a small hypothesis test: does class matter, does gender matter, does age matter? The answers pointed to which columns deserve to become model features and which problems (missing values, imbalance, skew) I would have to deal with first. This connects directly to the workflow from the previous lab: EDA is the data-understanding stage that decides how well every later stage can go.

The debugging experience taught me as much as the plots

My first real challenge was not statistical at all. The notebook failed on its very first code cell with a SyntaxError, and every later cell then threw a NameError because pd, plt and sns were never defined. My first instinct was to look at the last error, but the lesson was to read the first one, because later errors were just a cascade from the failed import cell. When I inspected the file, I found the root cause: the line breaks in the code cells had been lost, so entire cells were being read as one long line. In some cells a # comment then swallowed all the code after it, which meant the cell appeared to run but did nothing. Fixing it also exposed a second problem, a \n inside a print string that had been split across two lines. This was a good reminder that code can fail silently, and that checking your environment and inputs comes before analysis, just as checking data quality comes before modeling.

Insights from the analysis

The Embarked experiment was my biggest breakthrough. The raw countplot suggested Southampton was the deadliest port, but that was mostly because far more people boarded there. Comparing survival *rates* instead of counts told the real story (Cherbourg about 55%, Queenstown about 39%, Southampton about 34%). The crosstab then showed that Cherbourg had a larger share of 1st-class passengers, so port of embarkation is probably standing in for passenger class rather than causing survival. That was my first real encounter with confounding, and it made me more careful about reading a pattern as a cause.

The Fare boxplot taught me why distribution shape matters. Fares are heavily right-skewed, so a few very expensive tickets compress the boxes on a normal axis. Switching to a log scale made the difference between survivors and non-survivors readable, and it showed me why the median is a safer summary than the mean for skewed data. Similarly, the average age of about 29.7 told me almost nothing, while the histogram revealed a spike of surviving young children. That is the clearest answer I found to why summary statistics alone are not enough.

Connections and open questions

The overall survival rate is only around 38%, so the dataset is imbalanced. A model that predicted 'died' for everyone would already be right about 62% of the time, which means accuracy alone would be a misleading way to judge a model. I also noticed that Age and Cabin have many missing values, so I would need a deliberate strategy for them rather than just dropping rows. In the previous lab I compared different feature sets and chose the one that performed better; EDA is what would let me justify such a choice with evidence before training anything.

Some questions remain. Is the gender effect still strong inside each passenger class, or is some of it class again? Are passengers with missing Age systematically different from the rest, which would bias any conclusion about age? Answering these would need grouped plots and cross-tabulations, and I would like to try them next. Overall, my approach has shifted from 'run the cells and read the insight printed under each plot' to 'question what each plot can and cannot prove.'

Johna Matthews

Academic Reflective Journal

1. Introduction

The lab focused on Exploratory Data Analysis (EDA), an important stage in the data science process. EDA involves examining a dataset using statistics and visualizations to understand its structure, identify patterns, and recognize potential data-quality issues before building a machine learning model. The lab used the Titanic dataset to explore factors associated with passenger survival.

2. Description of the Experience

The lab began by introducing EDA and explaining the importance of understanding data before applying machine learning techniques. Using Python and Pandas, I examined the Titanic dataset with functions such as head() and info(). This allowed me to understand the variables, data types, and missing values in the dataset. Missing information was identified in the Age and Cabin columns.

I then used descriptive statistics with the describe() function. This provided information such as the mean, standard deviation, minimum, and maximum values. The analysis showed that the average passenger age was approximately 29.7 years and that the overall survival rate was approximately 38.4%.

The lab also introduced visual EDA using Matplotlib and Seaborn. Countplots were used to examine survival, passenger class, and gender. These visualizations showed that more passengers died than survived and that survival was strongly associated with passenger class and gender.

The lab also used histograms to examine the age distribution of passengers according to survival status.

3. Personal Reflection

At the beginning of the lab, I viewed EDA mainly as calculating statistics and looking at a dataset. However, the activities helped me understand that EDA is a broader process of asking questions and using different techniques to discover patterns in data.

I found the visualizations especially useful because they made relationships easier to understand. The lab helped me understand the importance of connecting technical analysis with interpretation. Rather than simply producing a graph, I need to consider what it shows and why the pattern may be important. This is particularly relevant when preparing data for machine learning because EDA can help identify variables that may be useful predictors.

The lab also connected with the theoretical concept of data preprocessing. Identifying missing values in the dataset demonstrates why data quality must be examined before a model is developed. I learned that incomplete or problematic data can affect later analysis and therefore needs to be considered during the early stages of a project.

One important critical-thinking lesson was that relationships observed through EDA should not automatically be interpreted as causal relationships. For example, although passenger class was associated with survival, the analysis alone cannot establish that passenger class directly caused survival. Other factors may also have influenced the results.

4. Improvements and Learning

The lab improved my technical skills in Python, particularly my ability to use Pandas for data inspection and descriptive statistics, and Seaborn and Matplotlib for visualization. I also became more comfortable interpreting graphs and using them to identify patterns.

A major area of personal growth was learning to approach a dataset systematically. In future projects, I will first examine the structure and quality of the data before beginning modeling. I will check for missing values, investigate numerical and categorical variables, and use appropriate visualizations to understand relationships.

I also recognize that I need more practice performing independent analysis. The lab included activities involving port of embarkation and fare, which provide opportunities to investigate additional relationships. Completing similar independent experiments in future work will help me become more confident in choosing appropriate visualizations and interpreting results without relying entirely on examples.

5. Conclusion

The EDA lab provided valuable practical experience in understanding and exploring data before machine learning. I learned how to inspect a dataset, identify missing values, calculate descriptive statistics, and use visualizations to investigate patterns. The most important lesson I learned was that effective data analysis requires both numerical and visual investigation. The Titanic dataset demonstrated how variables such as passenger class and gender can reveal important patterns in survival.

Jonathon Wilturner

Reflective Journal: Module 04 Lab, Exploratory Data Analysis

ITAI 1371 - Introduction to Machine Learning

How my view of EDA changed

Before this lab, I thought EDA was just making a few charts to make the notebook look nice. Now I see it as the step where you ask real questions about the data before you ever build a model. Every plot worked like a small test: does class matter, does gender matter, does age matter? The answers showed me which columns are worth using as features, and which problems, missing values, imbalance, skewed data, I need to fix first. This ties straight back to the last lab. EDA is the stage where you actually understand the data, and that understanding decides how well every stage after it goes.

Debugging taught me as much as the charts did

My first real problem had nothing to do with statistics. The notebook broke on the very first code cell with a SyntaxError, and every cell after that threw a NameError because pd, plt, and sns were never defined. At first I wanted to chase the last error I saw, but the real fix was going back to the first error, because everything after it was just a chain reaction from that one broken import cell. Digging into the file, I found the actual cause: the line breaks inside the code cells had gotten lost, so whole cells were reading as one giant line. In some cells, a # comment ended up swallowing all the code after it, so the cell looked like it ran but actually did nothing. Once I fixed that, I found a second issue, a \n inside a print string that had gotten split across two lines. That was a good reminder that code can fail quietly, and that you check your environment and

your inputs before you trust any analysis, same way you check data quality before you trust a model.

What the analysis actually showed me

The Embarked breakdown was the biggest eye-opener. The raw countplot made it look like Southampton was the deadliest port, but that was mostly just because way more people boarded there. Once I compared survival rates instead of raw counts, the real picture showed up: Cherbourg around 55%, Queenstown around 39%, Southampton around 34%. Then the crosstab showed Cherbourg had more 1st-class passengers than the other ports, so the port itself probably wasn't the real factor. Passenger class was likely driving it. That was my first real run-in with a confounding variable, and it made me a lot more careful about calling a pattern a cause.

The Fare boxplot taught me why the shape of your data matters. Fares are heavily skewed, so a handful of expensive tickets flatten everything else on a normal scale. Switching to a log scale made the gap between survivors and non-survivors actually visible, and it showed me why median beats mean when your data is skewed like that. Same thing with age. The average age, about 29.7, told me almost nothing on its own. The histogram is what showed a spike of young kids surviving. That's the clearest example I found of why summary numbers alone don't cut it.

Connections and questions I still have

Only about 38% of passengers survived, so this data is imbalanced. A model that just guessed "died" for every single passenger would still be right 62% of the time, which means accuracy by itself is a bad way to judge a model here. I also noticed Age and Cabin both have a lot of missing values, so I'd need an actual plan for handling that instead of just dropping rows. In the last lab, I

tested different feature sets and picked the stronger one. EDA is the part that actually lets you back up a decision like that with evidence before you train anything.

A few things I still want to dig into. Does the gender effect hold up inside each passenger class on its own, or is that also just class showing up again? Are passengers missing an Age value different in some systematic way from everyone else, which could throw off any conclusion drawn from age? Answering both would take grouped plots and more crosstabs, and that's my next move. Overall, my mindset shifted from "run the cells and read whatever insight gets printed under the plot" to "question what each plot can actually prove, and what it can't."